

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Weeks 33-34
24 August – 6 September 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



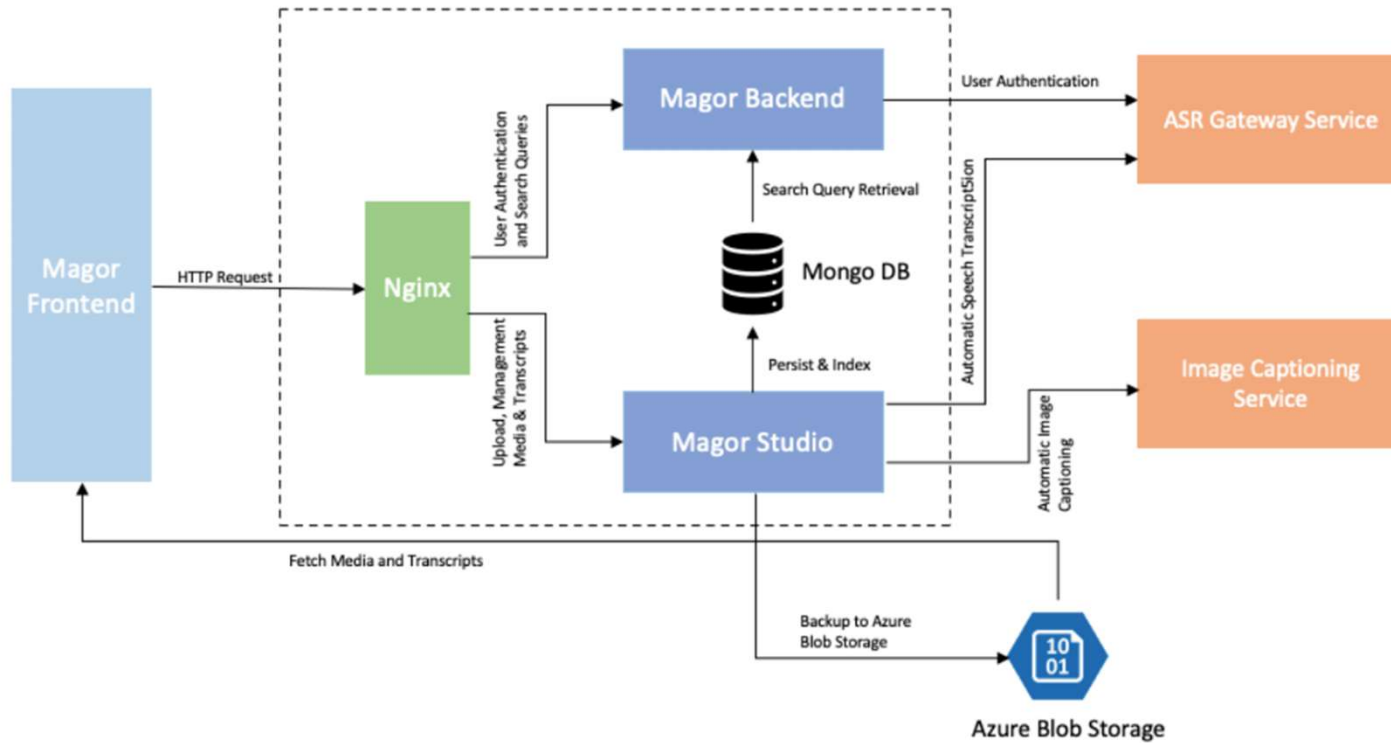
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work Done Over The Weeks:

- Fixed Timestamp click for 00:00
- Implemented Q&A history pagination
- Fixed unnecessary description hover effect when not truncated
- Frontend Node Upgrades:
 - Upgraded Node.js to version 24
 - Fixed error handling for refresh ASR requests status
 - WIP: Added CRACO configuration for missing browser replacements



Task 1 (Fixed Timestamp click at 00:00)

- No response when clicking timestamps at 00:00:

The screenshot displays the Magor video player interface. The top navigation bar is blue with the 'Magor' logo, a search bar, and a 'This recording' dropdown. Below the bar, the video title 'fyp faulty video.mp4' is shown. The video player itself shows a woman speaking with Chinese subtitles: '哥哥们在学怎么喂她' and 'They still hate each other'. The video progress bar indicates 6:42 / 8:50. To the right of the video player, there is a 'Transcript' tab and an 'Ask AI' button. Below these, a 'Summary' section is visible, containing a paragraph of text and a list of 'Key points' with bullet points. The 'Key points' include: 'Large family logistics', 'Shift to public education', 'Teaching financial awareness', and 'Response to a high-risk pregnancy'. At the bottom right, there is a 'Generate New Summary' button.

Magor Search This recording test

fyp faulty video.mp4

Transcript Ask AI

Summarize Q&A

Summary

The conversation explores the dynamics of a Singaporean family with eight children, touching on the challenges and joys of large-family life, financial and emotional adjustments, and how the siblings cope with a newborn who may have health issues. It highlights personal reflections on parenting, sibling relationships, and difficult decisions around prenatal testing and special needs (00:00-08:34).

Key points:

- **Large family logistics:** The family uses bulk shopping (25kg rice, 90 eggs monthly) and large appliances to manage daily life, reflecting the practical demands of feeding and caring for eight children (02:15-02:36).
- **Shift to public education:** After initially enrolling the first two children in costly private kindergarten (\$\$2000/month each), the parents switched to public preschool for all subsequent kids, prioritizing financial sustainability while noting the quality was comparable (02:51-03:02).
- **Teaching financial awareness:** The parents intentionally instill money-saving values in their children, explaining they can't always buy what they want, helping them understand financial limits (03:24-03:34).
- **Response to a high-risk pregnancy:** When prenatal tests suggested a 1 in 12 chance of health issues for their baby Eva (vs. 1 in thousands normally), the parents chose not to pursue further testing and decided to continue the pregnancy without hesitation (04:06-04:46).

0 matches They still hate each other The older ones I'm trying to teach them actually how to

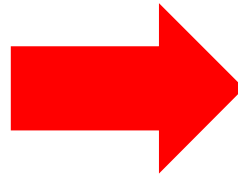
6:42 / 8:50

Generate New Summary

Task 1 (Fixed Timestamp click at 00:00)

Passing jump as 0 would return FALSE!

```
useEffect(() => {  
  if (media && jump) {  
    media.currentTime = jump / 1000  
    setSeek(jump)  
    jumpTo(null)  
  }  
  // eslint-disable-next-line react-hooks  
}, [jump])
```



Check for null instead of truthiness

```
useEffect(() => {  
  if (media && jump !== null) {  
    media.currentTime = jump / 1000  
    setSeek(jump)  
    jumpTo(null)  
  }  
  // eslint-disable-next-line react-hooks  
}, [jump])
```



Task 1 (Fixed Timestamp click at 00:00)

- Fixed timestamp click demo:

Magor

Search

This recording

test

fyp faulty video.mp4

现代

早

林哲安 (10岁)

1 2 3 4 5 6 7 8

0 matches

What?

2 9 10

8 4

0:01 / 8:50

Transcript

Ask AI

Edit

Summarize

Q&A

Summary

The conversation explores the dynamics of a large Singaporean family with eight children, discussing the challenges and joys of raising such a big family, financial and emotional adjustments, and how the children cope with the possibility of a new sibling having health issues. The parents reflect on their lifestyle, parenting decisions, and the strong bonds among siblings despite the chaos (00:00-08:35).

Key points:

- **Large family logistics:** The family uses oversized appliances and buys groceries in bulk, such as 25kg rice and 90 eggs monthly, to manage their large household efficiently (02:15-02:36).
- **Cost-saving education choice:** After initially enrolling the first two children in expensive private kindergarten, the parents switched all children to public kindergarten to save money, believing the education quality was sufficient (02:55-03:02).
- **Teaching financial awareness:** The parents intentionally teach their children to be mindful of spending and not expect everything they want, instilling values of frugality and gratitude (03:24-03:34).
- **Preparing kids for a high-risk birth:** When expecting a baby with potential health issues, the parents openly discussed possible financial sacrifices and even the baby's survival with the older children, who collectively decided to welcome the baby (04:57-05:57).

Hide

Generate New Summary



Task 2 (Implemented Q&A history pagination)

- Previous Q&A history loaded all entries:

The screenshot displays the Magor interface. At the top, there is a blue header bar with the name 'Magor', a search bar, and a dropdown menu set to 'This recording'. Below the header, the main content area features a video player with the title 'Indonesian stocks plunge more than 7 after MSCI flags risks'. The video player shows a news anchor reporting on the stock market. To the right of the video player, there is a 'Transcript' tab and an 'Ask AI' button. Below these, there is a 'Summarize' tab and a 'Q&A' tab. The 'Q&A' tab is active, showing a list of previous questions and answers. A red box highlights the 'Previous Q&A' section, which contains two entries. A red arrow points from the text 'Will load all previous Q&A' to the 'Previous Q&A' section. The interface also includes a 'Search' bar, a 'test' dropdown, and an 'Edit' button.

Magor Search This recording test

Indonesian stocks plunge more than 7 after MSCI flags risks

Transcript Ask AI

Summarize Q&A

Ask a question about the transcript... Ask

Previous Q&A

Q: Who are the main parties involved?

A: The main parties involved are MSCI, which issued a warning about Indonesia's market investability and potential downgrade (00:11-00:22); the Indonesian government and regulators, including Bank Indonesia (the central bank) and the Indonesia Stock Exchange (IDX) (03:10-03:37); and key political figures such as President Prabowo Subianto and his nephew, Thomas, nominated for deputy governor of Bank Indonesia (02:16-02:20). Investor confidence is also a central concern, with foreign investors highlighting issues around fiscal discipline and transparency (01:32-02:27).

Q: When did this occur?

A: The recording does not specify the exact date it occurred, but it references events such as the currency sliding since August 2025 (00:39-00:43), the resignation of finance minister Sri Mulyani Indrawati in September (00:47-00:51), and a nomination in January (00:58-01:02). It also mentions a May 2026 deadline for Indonesia to address MSCI's concerns (03:34-03:37). Based on these timestamps, the report appears to have been made in early 2026, likely before May.

Will load all previous Q&A



Task 2 (Implemented Q&A history pagination)

- Only a fixed number of Q&A exchanges are loaded at a time
- More will be retrieved as the user scrolls.

The screenshot displays the Magor application interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, and a 'This recording' dropdown menu. Below the header, the main content area features a video player on the left and a Q&A summary panel on the right. The video player shows a news broadcast titled 'ASIA Business Tonight' with a red overlay. The Q&A panel is titled 'Summary' and contains a paragraph of text about Indonesian stocks plunging nearly 9%, triggering a circuit breaker. Below the paragraph, there are three bullet points under the heading 'Key points:'. The first bullet point is 'Stock market plunge: Indonesian shares dropped nearly 9%, prompting a 30-minute trading halt, with the Jakarta composite closing down over 7% (00:00-00:11)'. The second bullet point is 'MSCI warning: MSCI flagged Indonesia's market instability, citing thin trading and ownership concentration, threatening a downgrade to frontier market status if reforms aren't made by May 2026 (00:11-00:26)'. The third bullet point is 'Fiscal and central bank concerns: Investor confidence is weakened by fiscal discipline worries and perceived erosion of Bank Indonesia's independence, especially after a surprise rate cut and political appointments (00:39-01:10)'. The fourth bullet point is 'Controversial appointment: The nomination of President Prabowo Subianto's nephew as deputy governor of Bank Indonesia raised concerns about institutional independence and governance (01:02-02:20)'. At the bottom of the Q&A panel, there is a blue button labeled 'Generate New Summary'.

Magor Search This recording test Edit

Indonesian stocks plunge more than 7 after MSCI flags risks

Transcript Ask AI

Summarize Q&A

Summary

Indonesian stocks plunged nearly 9%, triggering a circuit breaker, amid concerns over market instability and potential MSCI downgrade to frontier market status, with investor confidence shaken by issues around central bank independence and fiscal discipline (00:00-03:39).

Key points:

- **Stock market plunge:** Indonesian shares dropped nearly 9%, prompting a 30-minute trading halt, with the Jakarta composite closing down over 7% (00:00-00:11).
- **MSCI warning:** MSCI flagged Indonesia's market instability, citing thin trading and ownership concentration, threatening a downgrade to frontier market status if reforms aren't made by May 2026 (00:11-00:26).
- **Fiscal and central bank concerns:** Investor confidence is weakened by fiscal discipline worries and perceived erosion of Bank Indonesia's independence, especially after a surprise rate cut and political appointments (00:39-01:10).
- **Controversial appointment:** The nomination of President Prabowo Subianto's nephew as deputy governor of Bank Indonesia raised concerns about institutional independence and governance (01:02-02:20).

Hide

Generate New Summary

Task 3 (Removed unnecessary hover effect in description)

- Unnecessary hover effect on short descriptions:

Magor Search This recording test

cna flood.mp4 Edit

Hundreds of families await word on the fate of thousands still missing

KATHMANDU

NEPAL-TIBET DISASTER

Hundreds of families await word on the fate of thousands still missing

0 matches

Nothing is playing right now.

Transcript Ask AI Show options

speaker1 Among those still unaccounted for are nearly 100 Chinese nationals whose relatives are desperate for news, and CNA spoke with one man who traveled to Kathmandu in search of his wife.

speaker2 faces of the missing plastered on walls at relief camps and hospitals in the disaster hit district. As families search for their loved ones, among them Chinese national Lily Shah missing since a massive smartlight struck the Nepal Tibet border a week ago.

speaker2 Her husband, 38 year old Guan Do last spoke to her the night before disaster struck.

channel1_S3 就是很家常的一些话. 但是第二天中午我得知这边友谊士流之后, 应该是中国时间的11:30左右. 我再跟他联系就没有任何回应了.

speaker2 Miss Lee was working on a highway project in Nepal near China's



Task 3 (Removed unnecessary hover effect in description)

- Fixed Description Demo:

Magor

Search

This recording

test

cna flood.mp4

Edit

Transcript

Ask AI

Show options

KATHMANDU

NEPAL-TIBET DISASTER

Hundreds of families await word on the fate of thousands still missing

r flood disaster, seeks safer sites: Report Getting aid to Nepal flood survivors very difficu

0 matches

< >

Nothing is playing right now.

0:00 / 4:12

speaker1

Among those still unaccounted for are nearly 100 Chinese nationals whose relatives are desperate for news, and CNA spoke with one man who traveled to Kathmandu in search of his wife.

speaker2

faces of the missing plastered on walls at relief camps and hospitals in the disaster hit district. As families search for their loved ones, among them Chinese national Lily Shah missing since a massive smartlight struck the Nepal Tibet border a week ago.

speaker2

Her husband, 38 year old Guan Do last spoke to her the night before disaster struck.

channel1_S3

就是很家常的一些话. 但是第二天中午我得知这边友谊士流之后, 应该是中国时间的11:30左右. 我再跟他联系就没有任何回应了.

speaker2

Miss Lee was working on a highway project in Nepal near China's Jurong border checkpoint.



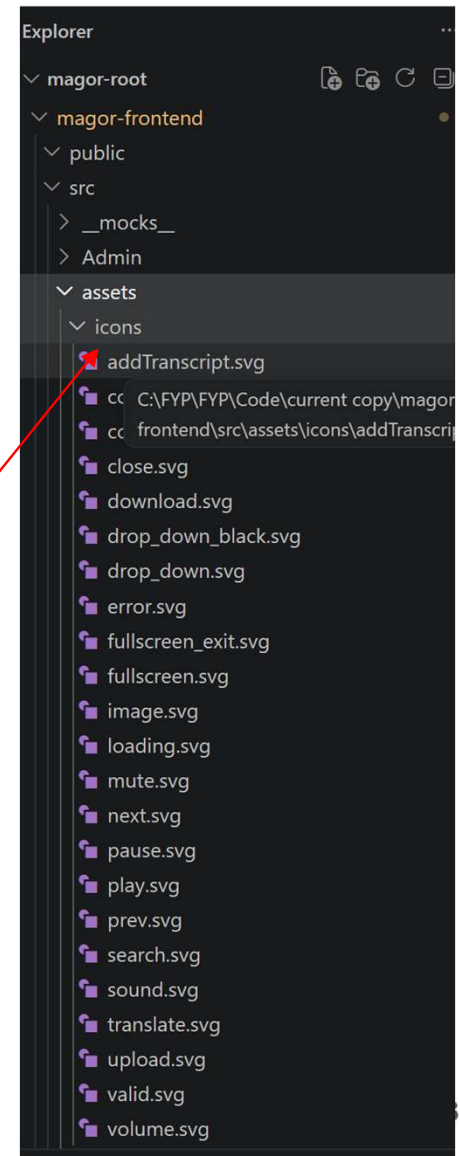
Task 4 (Frontend Node Upgrade)

- Frontend Node Upgrade to 24
 - Fixed SVG icon build errors
 - Webpack 5 could no longer resolve root-relative SVG paths
 - Moved 18 SVGs to src/assets/icons/ and updated their references

Upgraded
to Node
24-Alpine

```
1 FROM node:24-alpine
2
3 # Optional: needed if you have native deps
4 RUN apk add --no-cache python3 make g++
5
6 WORKDIR /usr/src/app
7
8 COPY package.json yarn.lock ./
9 RUN yarn install --frozen-lockfile
10
11 COPY . .
12
13 ENV PORT=3001
14
15 EXPOSE 3001
16
17 CMD ["yarn", "start"]
```

Moved ALL
SVGs to icons
folder



Task 4 (Frontend Node Upgrade)

- Error handling for refresh ASR requests status:

Magor Studio

test

Save

Basic Details

Transcripts

Captions

Publish

Auto Transcribe

Current Running Jobs

```
{\"status\": \"pending\", \"_id\": \"6a9a3852b8211f0023007245\", \"mediaId\": \"6a9a37e4b8211f0023007234\", \"audioType\": \"closetalk\", \"audioTrack\": \"single\", \"language\": \"trilingual\", \"customData\": {\"numSpk\": 1}, \"asrId\": \"6a9a385232a40bb248e273a2\", \"createdAt\": \"2026-09-04T03:17:38.674Z\", \"updatedAt\": \"2026-09-04T03:17:38.674Z\"}
```

Refresh

Send a new transcription job

Language

Abax Default (This model is customised for Singapore English Mandarin Malay)

Audio Type

Close Talk

Audio Track

Single Channel

Number of Speakers



Task 4 (Frontend Node Upgrade)

```
const handleRefreshAutoTranscribe = async (asrId) => {  
  toast  
    .promise(  
      fetchAutoTranscribeStatus(id, asrId),  
      {  
        loading: `Checking status for ASR Request: ${asrId}`,  
        success: (data) => `${data}`,  
        error: (err) => `${err}`,  
      },  
      {  
        error: {  
          duration: 3000,  
          icon: '🕒',  
        },  
      }  
    )  
    .then((data) => {  
      // console.log(data)  
      refreshMediaData()  
    })  
    .catch((err) => {  
      if (String(err) === ASR_IN_PROGRESS) return  
      console.error(err)  
    })  
}
```



Task 4 (Frontend Node Upgrade)

- Fixed ASR status refresh demo:

Magor Studio

test

Flood x Nepal x Disaster x CNA x

▶ 🔊 ⬇

Save

Basic Details Transcripts Captions Publish

Auto Transcribe

Current Running Jobs

```
{
  "status": "pending",
  "id": "6a9a417e30e1790023748d92",
  "mediaId": "6a9a37e4b8211f0023007234",
  "audioType": "closetalk",
  "audioTrack": "single",
  "language": "trilingual",
  "customData": {
    "numSpk": -1,
    "asrId": "6a9a417e32a40bb248e279a0",
    "createdAt": "2026-09-04T03:56:46.397Z",
    "updatedAt": "2026-09-04T03:56:46.397Z"
  }
}
```

Refresh

Send a new transcription job

Language

Abax Default (This model is customised for Singapore English Mandarin Malay)

Audio Type

Close Talk

Audio Track



Task 4 (Frontend Node Upgrade)

- XML libraries error:

The screenshot displays the Magor web application interface. At the top, there is a blue header bar with the 'Magor' logo, a search bar, and a 'This recording' dropdown menu. Below the header, the URL 'cna flood.mp4' is visible. The main content area is split into two sections. On the left is a video player showing a news broadcast from CNA about the Nepal-Tibet disaster. The video player has a progress bar at the bottom indicating '0:00 / 4:12'. On the right is a transcript panel with a 'Transcript' dropdown menu, 'Download', and 'Save' buttons. The transcript is organized into three segments, each with a timestamp and a channel label: 00:00:00 (channel1_S1), 00:00:13 (channel1_S2), and 00:00:30 (channel1_S2). The first segment's text is: 'Among those still unaccounted for are nearly 100 Chinese nationals whose relatives are desperate for news, and CNA spoke with one man who traveled to Kathmandu in search of his wife.' The second segment's text is: 'faces of the missing plastered on walls at relief camps and hospitals in the disaster hit district. As families search for their loved ones, among them Chinese national Lily Shah missing since a massive smartlight struck the Nepal Tibet border a week ago.' The third segment's text is: 'Her husband, 38 year old Guan Do last spoke to her the night before disaster struck.'



Task 4 (Frontend Node Upgrade)

WIP: XML browser replacements

- Added CRACO configuration for missing browser replacements
- Exploring newer versions or alternatives to replace the outdated dependencies

```
1 // CRA 5 runs webpack 5, which no longer provides Node modules or globals for the browser.  
2 const webpack = require('webpack')  
3  
4 module.exports = {  
5   webpack: {  
6     configure: (webpackConfig) => {  
7       // Modules dependency actually imports (xml2js require('timers'))  
8       webpackConfig.resolve.fallback = {  
9         ...webpackConfig.resolve.fallback,  
10        timers: require.resolve('timers-browserify'),  
11        buffer: require.resolve('buffer/'),  
12        process: require.resolve('process/browser'),  
13      }  
14  
15      // Globals used without being imported (xml-js: `json instanceof Buffer`)  
16      // the fallback above has no import to redirect. Rewrites them into real imports  
17      webpackConfig.plugins.push(  
18        new webpack.ProvidePlugin({  
19          Buffer: ['buffer', 'Buffer'],  
20          process: 'process/browser',  
21        })  
22      )  
23  
24      return webpackConfig  
25    },  
26  },  
27 }
```

Plans for Next Update

- Resolve XML dependency issues
- Fix transcript selection in “Show Options” panel
- Upgrade Studio/Backend Node Version and resolve dependency issues

